

Gaurav Budhwani

Senior Undergraduate · Modelling & Optimization · Machine Learning & Data Science

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EDUCATION

Indian Institute of Technology, Gandhinagar [\[Transcript\]](#) CPI: 8.46/10
B.Tech. (Dual Majors) - Chemical and Computer Science Engineering 2022 – Present

INTERNSHIPS

Research Scholar - PMTL Lab, University of Miami Jun '25 – Jul '25
Advisor — Prof. Yang Wang, Prof. Pratim Biswas, University of Miami, Floria, USA

- Built a scalable ML pipeline for **large-scale sensor data**, benchmarking **5 classifiers** and achieving **ROC-AUC up to 0.98** through **feature engineering**, **hyperparameter tuning**, and **effective class imbalance handling**.
- Enhanced prediction reliability using **SMOTE**, **feature importance**, and **PDPs**, delivering **interpretable models** validated against **established scientific literature** and **real-world observations** for **robust deployment**.

AEH Intern in Cloud Capability, Accenture May '26 – Jul '26
Delivery Lead — Nitin Dabli, Sr. Vice President, Azure Cloud & DevOps

- Researched **emerging AI development tools**, identifying strategic solutions for **SDLC optimization**, developer productivity, and engineering efficiency.
- Architected a proof-of-concept **AI-driven platform engineering agent** using **LangGraph**, **FastAPI**, and **Azure OpenAI** to **automate infrastructure provisioning** and generate **security-compliant Terraform plans**.

PUBLICATIONS

- Hao, W., Mehra, M., **Budhwani, G.**, Chakraborty, T. C., Mei, F., Wang, Y. (2026). Employing machine learning for new particle formation identification and mechanistic analysis: Insights from a six-year observational study in the Southern Great Plains. Journal of Geophysical Research: Atmospheres, 131, e2024JD043116. [Journal Link](#)

PROJECTS

Dynamic Modeling of a Multi-Zone Residential House in Delhi Jan '26 – May '26
Project Course | Advisor — Prof. Hari S. Ganesh, IIT Gandhinagar | [\[Poster\]](#)

- Developed a **3-zone RC thermal model** of a Delhi residence, capturing **inter-zone heat transfer**, wall dynamics, solar gains, internal heat gains, and **ventilation/infiltration**.
- Integrated **Delhi weather data** with dynamic **occupancy**, **appliance**, **cooking**, **window**, **ventilation**, and **AC schedules** to simulate realistic transient indoor temperature and heat-load profiles.
- Formulated a **control-ready dynamic model** for future **Energy-focused Model Predictive Control (MPC)**, targeting thermal comfort and indoor air quality optimization.

TCP Congestion Control Simulator Aug '25 – Nov '25
Advisor — Prof. Sameer G. Kulkarni, IIT Gandhinagar | [\[Project Link\]](#)

- Developed an interactive **TCP congestion-control visualizer** with a **Python–Flask backend** and **React frontend**, enabling real-time plots of cwnd, throughput, and packet loss for **TCP Reno**, **CUBIC**, and **BBR**.
- Implemented a custom **discrete-packet simulation engine** and an **ns-3 packet-level backend**, providing support for single-flow and multi-flow scenarios with programmable bandwidth and routing.
- Engineered **Custom-topology support** (parallel, series, triangle, mesh) with dynamic route selection to visualize congestion.

API Gateway Core Nov '25 – Dec '25
[Independent Project](#) | [\[Project Link\]](#)

- Designed and implemented a **production-style API Gateway core** in **C++17** using a **middleware-based request processing pipeline** (Chain of Responsibility) supporting authentication, rate limiting, and logging.
- Developed a **concurrent request execution model** using a custom **thread pool** and **request-scoped context objects**, ensuring thread safety, isolation of per-request state, and correct behavior under parallel workloads.
- Implemented **prefix-based routing** and clean **backend service abstractions**, enabling modular service integration and graceful handling of failure scenarios including **401 (Unauthorized)**, **404 (Not Found)**, and **429 (Too Many Requests)**.

VisuAlgo CSP Solver for N-Queens and KenKen Oct '25 – Nov '25
Advisor - Prof. Neeldhara Misra, IIT Gandhinagar | [\[Project Link\]](#)

- Developed an interactive web app to visualize Constraint Satisfaction Problem (CSP) algorithms using **React** and

TypeScript.

- Implemented complex solving strategies including **Backtracking, Forward Checking, and Arc Consistency (AC-3)** to demonstrate constraint propagation in real-time.
- Designed a dynamic user interface with **React Router** that features interactive puzzle generation, step-by-step solution playback, and visual feedback for constraint violations.

Human Activity Recognition (HAR) Analysis

Aug '25 – Sep '25

Advisor - Prof. Nipun Batra, IIT Gandhinagar | [\[Project Link\]](#)

- Analyzed the **UCI-HAR dataset** consisting of time-series sensor data from 30 subjects performing 6 daily activities, and applied **Principal Component Analysis (PCA)** to reduce dimensionality and visualize activity clusters.
- Utilized the **TSFEL library** to extract diverse statistical, temporal, and spectral features, and trained a **Decision Tree model** on the featurized dataset to classify human activities.
- Collected real-world activity data using the **Physics Toolbox Suite** app for model validation, achieving **70% precision** and **70% accuracy**, demonstrating the feasibility of deploying the pipeline beyond benchmark datasets.

Modeling and Parameter Estimation for Pyrolysis of Wood Particles

Mar '25 – Apr '25

Advisor - Prof. Karthik S. Pushpavanam, IIT Gandhinagar | [\[Project Link\]](#)

- Developed a **multiphysics simulation in COMSOL** for wood pyrolysis under inert conditions, coupling **heat transfer, mass transport, and chemical kinetics** using anisotropic material properties.
- Conducted **parameter estimation using nonlinear least squares** to optimize kinetic and thermal parameters, improving alignment with experimental temperature and mass loss data.
- Performed **parametric studies** varying porosity, particle size, reaction kinetics, and ambient temperature to analyze their impact on thermal degradation, volatile release, and internal transport mechanisms.

Process Design and Simulation for Cumene Production

Feb '25 – Apr '25

Advisor - Prof. Hari S. Ganesh, IIT Gandhinagar | [\[Project Link\]](#)

- Designed a complete **cumene production process in Aspen Plus** and **MATLAB**, integrating reaction, separation, and heat recovery units for sustainable and optimized performance.
- Modeled a **plug flow reactor** with primary and secondary reactions using **mass and energy balances**, and validated the results against Aspen simulations with **<2% error** for key components.
- Performed **multi-variable optimization using MATLAB's fmincon**, achieving an optimal benzene-to-propylene ratio and inlet temperature that **maximized cumene yield** while minimizing side-product formation.

Optimization of Heat Loss and Boundary Layer Thickness

Oct '24 – Nov '24

Advisor - Prof. Hari S. Ganesh, IIT Gandhinagar | [\[Project Link\]](#)

- Modeled and optimized **thermal boundary layer thickness** and **heat loss** using **Pyomo** and **MATLAB**, enhancing **convective heat transfer efficiency** for laminar airflow over a heated plate.
- Analyzed **trade-offs** between **heat dissipation** and **boundary layer** control under constrained flow conditions, demonstrating **optimal configurations** for improving thermal performance in engineering applications.

Saffman-Taylor Instability Using Hele-Shaw Cells

Oct '24 – Nov '24

Advisor - Prof. Kartik Subramaniam, IIT Gandhinagar | [\[Project Link\]](#)

- Simulated **viscous fingering patterns** through **numerical modeling** (Darcy's Law, Diffusion-Limited Aggregation) to analyze the Saffman-Taylor instability, emphasizing computational approaches at the chemical-engineering interface.
- Designed and conducted controlled **experiments with Hele-Shaw cells** to validate simulations, investigating the interplay of viscosity, surface tension, and flow dynamics for interfacial instability.

Quiz App as Attendance Taking App

May '24 – Jun '24

Advisor - Prof. Balgopal Komrath, IIT Gandhinagar | [\[Project Link\]](#)

- Designed and deployed a **web-based quiz** and attendance system using **Flask, JavaScript, Jinja2, and SQLite** to support **real-time quizzes** and **automated attendance** tracking.
- Implemented **geolocation validation, randomized number generation, and admin-approval workflows** to prevent cheating and enable secure session management with **automated reporting**.

Helical Fins in a Double Heat Exchanger

Jan '24 – May '24

Advisor - Prof. Biswajit Saha, IIT Gandhinagar | [\[Project Link\]](#)

- Conducted comprehensive calculations to determine the **heat transfer coefficient of circular finned pipes**, taking into account various parameters such as fluid properties, flow rates, and temperature gradients.
- Evaluated the **thermal efficiency** and **effectiveness** of the fin by analyzing its ability to enhance heat transfer and improve overall system performance.
- Developed a detailed **comparative analysis** between the circular finned pipe and a standard double heat exchanger.

Transient Thermal Analysis of a Car Brake System

Aug '23 – Nov '23

Advisor - Prof. Dilip Srinivas Sundaram & Prof. Akshaa Vatwani | [\[Project Link\]](#)

- Executed detailed **numerical simulations** to assess the transient thermal behavior of a car brake system during braking, **analyzing heat distribution** and **temperature changes** over time to evaluate system performance.
- Utilized **advanced computational tools** and **finite element analysis (FEA)** to model the heat distribution and dissipation patterns across brake components, including the rotor, pads, and calipers.

Increasing Efficiency of Rankine Cycle

Oct '23 – Nov '23

Advisor - Prof. Atul Bhargav, IIT Gandhinagar | [\[Project Link\]](#)

- Implemented **thermodynamic analysis** and **simulation techniques** to identify and **adjust key cycle parameters**, such as turbine inlet temperature and condenser pressure, to maximize thermal efficiency and overall power output.
- Developed **graphical representations** to illustrate the relationship between cycle adjustments and performance metrics, providing clear visual insights into how different parameters influence efficiency and steam quality.

Analysis of Datasets using Probability, ML & Stats

Jan '23 – Apr '23

Advisor - Prof. Shanmugathan Raman, IIT Gandhinagar | [\[Project Link\]](#)

- Conducted analysis of Airbnb hosts' data, implementing comprehensive **data cleaning** to ensure dataset accuracy. Extracted valuable insights into host performance and guest interactions.
- Applied **advanced statistical techniques**, including **correlation analysis**, **regression** models, and **clustering**, to uncover significant trends related to pricing, location, and guest reviews.

Personal Portfolio Website

May '24 – Jun '24

Self Initiated | [\[Website Link\]](#)

- Designed and developed a personal **portfolio website** using **HTML**, **CSS**, and **JavaScript** to showcase projects, skills, and achievements, featuring a **responsive design** for optimal viewing on various devices.
- Conceptualised **dynamic elements** and **user-friendly navigation**, including interactive project galleries, animated elements, and contact forms to enhance user engagement and experience.

Developing Games and Puzzles with C and C++ Using DSA

Jan '24 – Apr '24

Self Initiated | [\[Website Link\]](#)

- Designed and implemented **Connect4** and **Up-it-Up** games utilizing optimal move strategies and **graph traversal algorithms** for strategic gameplay and decision-making.
- Developed **Sudoku Solver** and **2x2x2 Rubik's Cube Solver**, employing **advanced algorithms** to enhance solving efficiency and optimal move calculations.

TECHNICAL SKILLS

- **Languages:** Python, C, C++, MATLAB, HTML, CSS, Javascript, SQL, Rust.
- **Tools:** Git, LaTeX, Quarto, GitHub, Adobe Illustrator, Arduino IDE, Autodesk Inventor, Tableau, Data Modelling.
- **Libraries:** PyTorch, NumPy, Pandas, Scikit-learn, SciPy, Plotly, Seaborn, Flask, SQLite, SMOTE, XGBoost, LightGBM.

ACHIEVEMENTS

- Received the **Academic Excellence Award** for achieving the highest CPI in AY 2023-24.
- Felicitated with **Dean's List Award** IITGN for **Semester II** for excellent academic performance.
- Awarded **Academic Research Ranking 2** (out of 50 students) for the Academic Year 2024-2025, recommended for institute-led academic internship and exchange programmes.
- Secured a grade of **A+** (11/10) for outstanding performance in the course of Numerical Methods by successfully doing the project and simulation on "**Transient Thermal Analysis of a Car Brake System.**"
- Completed **Deloitte Australia Data Analytics** Virtual Internship (Forage, Jun 2025), involving **Tableau dashboards**, **Excel-based classification**, and **business insight generation**. [\[Certificate\]](#)
- Earned the Specialist rank on Codeforces, highlighting strong competitive programming skills.

RELEVANT COURSES

Computer Science Courses

- ES 336: Computer Organisation Architecture
- CS 327: Compilers
- CS 330: Operating Systems
- CS 331: Computer Networks
- CS 431: Computer and Network Security
- CS 202: Software Tools & Techniques for CSE
- ES 242: Data Structures and Algorithms I
- ES 204: Digital Systems
- ES 214: Discrete Mathematics
- CS 201: Theory of Computing

AI / ML Courses

- ES 335: Machine Learning
- CS 329: Foundations of AI: Multiagent Systems

Chemical Engineering Courses

- ES 211: Thermodynamics
- CL 201: Chemical Process Calculations
- CL 202: Chemical Engineering Thermodynamics
- CL 203: Process Fluid Mechanics
- CL 204: Heat Transfer
- CL 205: Chemical Reaction Engineering I
- CL 313: Chemical Reaction Engineering II
- CL 314: Separation Processes I
- CL 315: Process Dynamics and Control
- CL 326: Integrated Chemical Engineering Lab I
- CL 316: Separation Processes II
- CL 317: Process Synthesis, Design & Simulation
- CL 325: Transport Phenomena
- CL 327: Integrated Chemical Engineering Lab II
- ES 604: Engineering Optimization
- ES 617: Design of Experiments
- CL 427: Formulation Science and Engineering
- CL 629: Fundamentals of Aerosol Science
- CL 633: Applications of AI in ChE

Mathematics, Statistics & Electrical Courses

- ES 114: Probability, Statistics, and Data Visualization
- MA 103: Calculus of Single Variable and Linear Algebra
- MA 203: Numerical Methods
- MA 204: Introduction to Partial Differential Equations
- MA 104: Ordinary Differential Equations
- ES 116: Principles and Applications of Electrical Engineering

TEACHING EXPERIENCE

CL 317 Process Synthesis Design and Simulation

Jan '26 – May '26

Prof. Hari S. Ganesh

- Conducted laboratory sessions in coordination with the course professor, guiding students through simulations, troubleshooting procedures, and reinforcing core concepts.
- Designed and evaluated assignments while mentoring laboratory project components, providing technical guidance and feedback throughout project development.

ES 604 Engineering Optimization

Aug '25 – Nov '25

Prof. Hari S. Ganesh

- Facilitated weekly doubt-clearing sessions and assignment discussions, assisting students with optimization concepts and problem-solving approaches.
- Evaluated course projects and provided constructive feedback, ensuring alignment with course objectives and methodologies.

POSITIONS OF RESPONSIBILITY

Design Team Member, Amalthea '23

Dec '22 – Feb '24

(Annual Technical Summit of IIT Gandhinagar)

- Created engaging social media designs for diverse platforms, utilizing Adobe Illustrator, Canva, and other graphic design software.
- Developed and executed visually compelling social media content for various platforms using Adobe Illustrator, Canva, and other graphic design tools.

Team Member, Vinteo

Jul '24 – May '25

(Film Making Club of IIT Gandhinagar)

- Contributed to a short film's production, handling concept development, and filming. Ensured smooth execution.

Team Member, MAPRC

Jul '24 – May '25

(Media and Public Relations Committee of IIT Gandhinagar)

- Designer for "ETHEREAL," IIT Gandhinagar's official monthly magazine, crafting visuals to showcase campus life, student achievements, events, and engaging stories.